Black holes & gravitational waves – Exam 2025

Guidelines

- This is a take-home exam. It will be graded. It will count for 60% of the total grade. Please hand-in your answers before **Noon Friday 20 June 2025**.
- Answers are handed digitally through e-mail to:
  maarten.meent@nbi.ku.dk & jose.ezquiaga@nbi.ku.dk.
  Please name all files: *LastName_FirstName_exam_2025.pdf*
- You must do the exam individually
- You are allowed to use computer algebra software (e.g. Mathematica) and coding in general (e.g. jupyter notebook of pyhton) to aid with your computations. However, if you do, please include your notebooks with your answer.
- Please make sure that your answers (and any supplemental material such as code) are comprehensible!

Question 1. Alice and Bob's black hole adventure

Alice and Bob are playing near a Kerr black hole with spin $a = 0.7M$, and have gotten themselves in a bit of bind.

In a desperate attempt to save themselves they push away from each other at a spacetime position with Boyer-Lindquist coordinates

$$\begin{aligned} t &= 0, \\ r &= 3M, \\ \cos\theta &= 0.8 \\ \phi &= 0. \end{aligned}$$

Bob heads-off with 4-velocity

$$\begin{aligned} \frac{dt}{d\tau} &= 4.65, \\ \frac{dr}{d\tau} &= -1.6, \\ \frac{d\theta}{d\tau} &= 0 \\ \frac{d\phi}{d\tau} &= 0.3 \end{aligned}$$

and Alice with 4-velocity

$$\frac{dt}{d\tau} = 3.113,$$
$$\frac{dr}{d\tau} = 0,$$
$$\frac{d\theta}{d\tau} = 0$$
$$\frac{d\phi}{d\tau} = 1.$$

a. Calculate the constants of motion μ, $\mathcal{E}$, $\mathcal{L}$, $\mathcal{Q}$ for Alice and Bob.
b. How many real roots does the radial potential P_r have for Alice and Bob?
c. What kind of geodesics will Alice and Bob follow.
d. Draw the Penrose diagram for the maximally extended Kerr spacetime, and sketch the trajectories of Alice and Bob.
e. Will Alice and Bob ever see each other again?

Question 2: Quasi-monochromatic gravitational wave sources

During the course we have studied the gravitational waves (GWs) produced during the inspiral phase of a compact binary coalescence. At leading order, the amplitude of the GW scales as:

$$h(t) = \frac{4}{d_L}\left(\frac{G\mathcal{M}_z}{c^2}\right)^{5/3}\left(\frac{\pi f_{\rm gw}(t)}{c}\right)^{2/3}\cos(\Phi(t))\,,$$

where d_L is the luminosity distance, c the speed of light and G Newton's constant. Note that we have omitted here the polarization dependence of the strain on the inclination angle. The phase is determined by the frequency evolution

$$\Phi(t) = 2\pi\int dt f_{\rm gw}(t)\,.$$

Recall that $\mathcal{M}_z = (1+z)\mathcal{M}_c$ is the redshifted chirp mass and $f_{\rm gw}(t)$ the frequency evolution measured at the detector, which can be computed through the energy loss of the orbit in GWs:

$$\dot{f}_{\rm gw} = \frac{96}{5}\pi^{8/3}\left(\frac{G\mathcal{M}_z}{c^3}\right)^{5/3} f_{\rm gw}^{11/3}\,.$$

Depending on the mass of the binary and the frequency range of observation, a given signal can be in different regimes. When for a given observing time $T_{\rm obs}$ the frequency evolves little compared to the signal itself,

$$\dot{f}\cdot T_{\rm obs} \ll f\,,$$

the signal is said to be *quasi-monochromatic*.

To obtain some of the results of this exercise it might be useful to utilize the following approximate integral relations

$$\int_0^T dt\, t^n \cos^2[Bt] \approx \int_0^T dt\, t^n \sin^2[Bt] \approx \frac{1}{2(n+1)} t^{n+1}\,,$$
$$\int_0^T dt\, t^n \cos[Bt] \approx \int_0^T dt\, t^n \sin[Bt] \approx 0\,,$$

when the range of integration T is much longer than the period of the integrand. Here n is an integer and B a constant.

1. Show that for a quasi-monochromatic source, the GW signal can be written at leading order as
$$h(t) = A\cos\left[2\pi\left(ft + \frac{1}{2}\dot{f}t^2\right) + \phi_0\right]\,,$$
where A and ϕ_0 are constants.

   For a binary neutron star of component masses equal to $1.4M_\odot$ that is observed for a year, what is the frequency range in which the quasi-monochromatic approximation will be valid? How does this condition scales with the time of observation and chirp mass?

2. The number of sources emitting GWs with a frequency larger than f is given by
$$N_{\rm gw}(>f) \sim RV_c\Delta t_{\rm merge}(f)\,, \tag{0.0.1}$$
where R is the comoving merger rate, V_c is the comoving volume and $\Delta t_{\rm merge}$ the time to merger for a signal with initial frequency f.

   How long it takes to merge a binary neutron star with an initial frequency of 1mHz? How does the number of events scales with the initial frequency?

   Given that from the latest GW catalog of LIGO–Virgo–KAGRA we know that the comoving merger rate of binary neutron stars is within 10–1700Gpc^{-3}yr^{-1}, and that the number density of Milky-way like galaxies is about $n_{\rm MW} \sim 0.01{\rm Mpc}^{-3}$, what would be the number of sources that we should expect to be emitting GW with frequencies larger than 1mHz in our own galaxy?

3. We want to compute the signal-to-noise ratio (SNR) of a quasi-monochromatic signal. To that end, first demonstrate that the noise-weighted inner product $(a|b)$ of two real functions $a(t)$ and $b(t)$ defined as
$$(a|b) = 4{\rm Re}\int_0^\infty \frac{\tilde{a}^*(f)b(f)}{S_n(f)}df$$
can be equivalently written in time domain as
$$(a|b) = \int_{-\infty}^{\infty} dt\hat{a}(t)\hat{b}(t)\,,$$

where $\hat{a}(t)$ is the inverse Fourier transform of the frequency domain noise-weighted signal $\tilde{\hat{a}} = \tilde{a}(f)/\sqrt{S_n(f)/2}$.

Then, show that for a quasi-monochromatic, the SNR accumulated during a time of observation $T_{\rm obs}$ at a frequency f is

$$\rho = \sqrt{(h|h)} \simeq \frac{A}{\sqrt{S_n(f)}} T_{\rm obs}^{1/2} \,.$$

4. Using the quasi-monochromatic *time* domain strain that you have just derived, compute the Fisher matrix

$$\Gamma_{ij} = (\partial_i h | \partial_j h) \,,$$

where $\partial_i = \partial/\partial\theta^i$ indicates a derivative in the parameter space $\vec{\theta} = \{A, f, \dot{f}, \phi_0\}$.

How do the errors in the parameters $\{A, f, \dot{f}, \phi_0\}$ scale with the observing time?[1] Which of them will be better constrained in a longer observing time?

5. The future Laser Interferometer Space Antenna (LISA) is an space-based detector that will operate in the milliHertz range thanks to its million-kilometer-long arms. Its power spectral density is expected to reach $S_n(f) \sim 10^{-40}\text{Hz}^{-1}$ at 5mHz. If we focus again on binary neutron star sources, do expect to detect them if they are within the galaxy?

What is the expected precision in the measurement of each parameter after 4 and 10 years of LISA observations?

6. If the GWs from a neutron star inspiral in the LISA band were to encounter a compact object such a stellar-mass black hole before arriving to the detector, will the signal be affected by gravitational lensing?

[1] Remember that you are allowed to invert the Fisher matrix with Mathematica, for example.